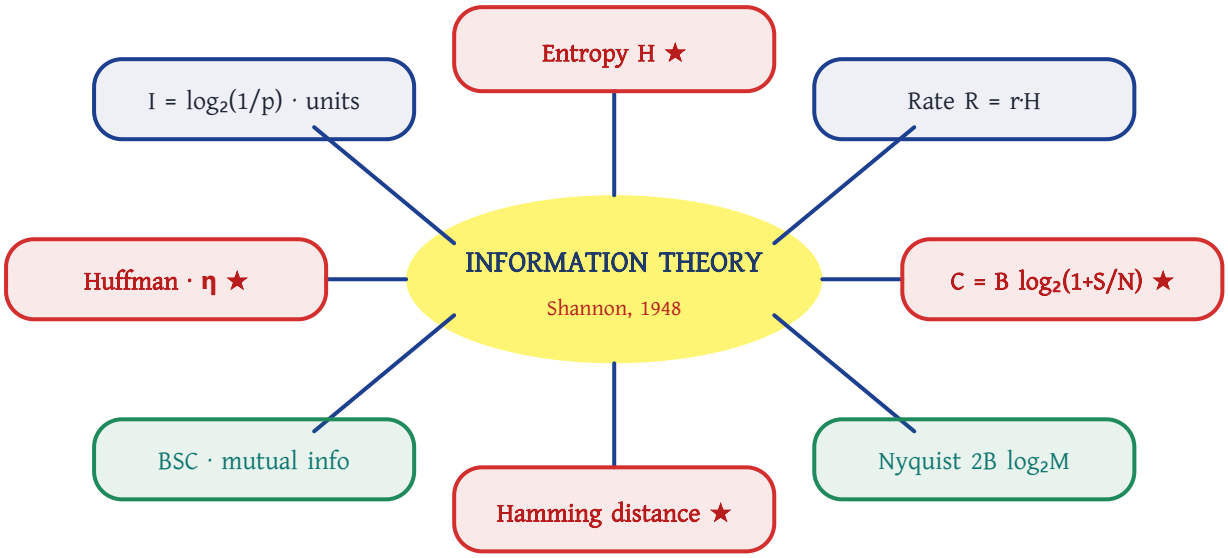
Information Theory

Entropy · Rate · Huffman · Shannon · Capacity: exam-ready in 11 sheets



↑ the whole topic on one page: red = high-yield, green = method / applied

Syllabus coverage

Communication Theory: "**Information theory: entropy, information rate, Shannon's theorem, channel capacity**". Also: source coding (Shannon–Fano, Huffman), the BSC, the Nyquist limit and error-control basics. Sampling is in the *Spectral Analysis & Sampling* notes; PCM, modulation and multiplexing in the *Modulation, Pulse Signals & Multiple Access* notes.

High-yield list

① $I = \log_2(1/p)$
② $H_{max} = \log_2 M$
③ $R = r \cdot H$
④ Huffman, $\eta = H/L$
⑤ $C = B \log_2(1 + S/N)$
⑥ limit −1.6 dB
⑦ d_{min} rules

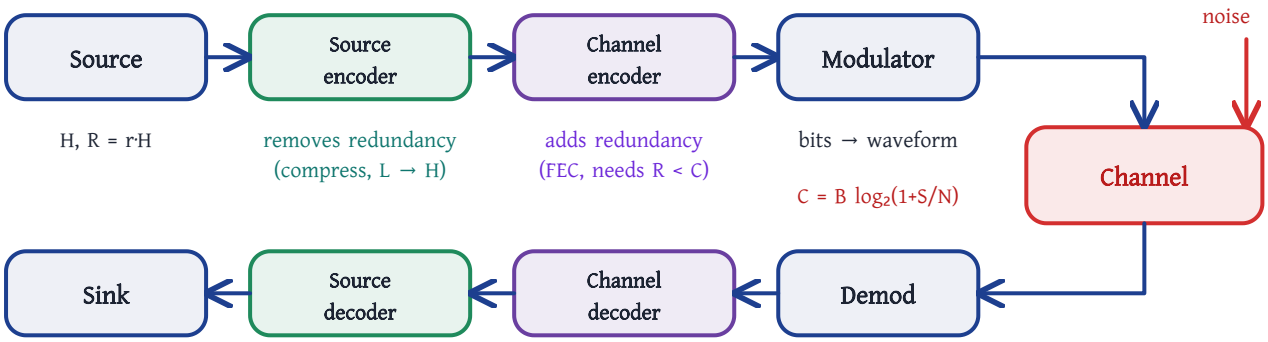
Legend: **yellow** = key term · **red circle** = must-memorise fact · **purple box** = memory trick · **red box** = exam trap · **green** = worked example

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✓ System model, information … p. 2
✓ Entropy & information rate … p. 3
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1 The Digital Communication System

Definition: Information theory is the mathematics of **measuring, compressing and reliably transmitting** information. Founded by Claude E. Shannon in "A Mathematical Theory of Communication" (Bell System Technical Journal, **1948**): the "father of information theory".



Shannon's model: source coding squeezes the redundancy out, channel coding puts controlled redundancy back in

Memory trick: "**S**ource coding **S**hrinks, **C**hannel coding **C**overs." Source coding = compression (Huffman, ZIP: L, η); channel coding = protection (parity, Hamming, CRC: code rate k/n, d_{min}).

2 Measure of Information

Self-information: A message of probability p carries $\mathbf{I = \log_2(1/p) = -\log_2 p}$ **bits**. The **less likely** the message, the **more information** it carries ("sun rose today" ≈ 0 bits; "earthquake in Dhaka" = many bits).

- p = 1 (certain event) → **I = 0**; always I ≥ 0
- $p_1 < p_2 \rightarrow I_1 > I_2$
- Independent messages: $\mathbf{I(x_1x_2) = I(x_1) + I(x_2)}$ (log turns × into +)
- Binary digit with p = ½ → exactly **1 bit**

Log base	Unit	= bits
2	**bit** (shannon)	1
e	**nat**	1.443
10	**decit / Hartley**	3.322

Worked example ✍️: p = 1/8 → I = $\log_2 8$ = **3 bits**. A card from a 52-card deck: $\log_2 52$ = **5.70 bits**; "it is an ace" (p = 4/52 = 1/13): $\log_2 13$ = **3.70 bits**.

Exam trap: **Baud** is a unit of **symbol rate**, not of information. And $\log_2 x = 3.322 \log_{10} x$: keep that conversion for calculator-free MCQs.

3 Entropy H: Average Information per Symbol

Definition: For a discrete memoryless source (DMS) with M symbols of probabilities $p_1 \dots p_M$: $H(X) = \Sigma\ p_i \log_2(1/p_i)$ bits/symbol. Entropy = **average self-information** = average uncertainty removed per symbol = the **limit of lossless compression**.

Bounds:

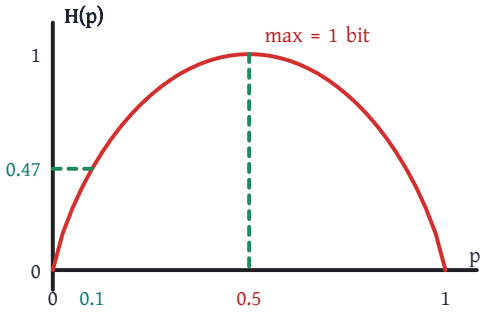
$0 \le H \le \log_2 M$

H = 0 → one symbol certain (p = 1)

$H = \log_2 M$ → all **equiprobable** (p = 1/M)

n-th extension: $H(X^n) = n \cdot H(X)$

Binary source: H(p) max **1 bit at p = 0.5**



binary entropy function

Source	Probabilities	H (bits/symbol)
Fair coin	½, ½	**1**
Biased coin	0.9, 0.1	0.469
4 equal symbols	¼ each	**2** = $\log_2 4$
Dyadic source	½, ¼, ⅛, ⅛	**1.75**
Fair die	1/6 each	2.585 = $\log_2 6$
English letters	letter frequencies	≈ 4.1 (max $\log_2 26$ = 4.70)

Worked example ✍️: p = ½, ¼, ⅛, ⅛: H = ½(1) + ¼(2) + ⅛(3) + ⅛(3) = 0.5 + 0.5 + 0.375 + 0.375 = **1.75 bits/symbol** (< $\log_2 4$ = 2, since not equiprobable). Source redundancy = 1 − H/H_{max} = 1 − 1.75/2 = 12.5%.

4 Information Rate R

Definition: Source emitting r symbols/s: **R = r · H** bits/s.

Worked example ✍️: A 4 kHz signal sampled at the Nyquist rate (r = 8000 samples/s), quantized to 4 levels with p = ⅛, ⅜, ⅜, ⅛: H = 2 × ⅛ × 3 + 2 × ⅜ × $\log_2(8/3)$ = 0.75 + 1.061 = **1.811 bits** → R = 8000 × 1.811 ≈ **14.5 kbit/s** (equiprobable levels: 16 kbit/s).

Memory trick: "**Equal → log, else less**": any imbalance lowers H.

Exam trap: H = 2 bits/symbol, r = 2000/s → R = **4000 bit/s**, not 1000.

5 Source Coding: Length, Efficiency, Kraft

Goal: Make the **average codeword length L** as small as possible without losing information: **short codes for frequent symbols** (the Morse idea).

Formulas:

Average length: $L = \Sigma\, p_i\, l_i$ bits/symbol

Efficiency: $\eta = H / L$ (binary code)

Redundancy: $\gamma = 1 - \eta$

D-ary code: $\eta = H / (L \log_2 D)$

Kraft (prefix): $\Sigma\, 2^{-l_i} \le 1$

Fixed-length code: $l = \lceil \log_2 M \rceil$ bits

Shannon's 1st theorem (source coding): For any uniquely decodable code $L \ge H$; a prefix code with $H \le L < H + 1$ always exists, and coding long blocks of symbols brings L/symbol as close to H as we like. $\eta = 100\%$ only when every p_i is a power of ½ (dyadic).

Code	A	B	C	D	Verdict
I	0	10	110	111	**prefix (instantaneous)**: no codeword starts another
II	0	01	011	111	not prefix (0 starts 01), still uniquely decodable
III	0	1	01	10	**ambiguous**: "01" = AB or C; Kraft sum = 1.5 > 1

6 Shannon–Fano Coding (top-down)

1. List the symbols in **decreasing** probability
2. **Split** the list into two parts with totals as **nearly equal** as possible; give 0 to the upper part, 1 to the lower
3. Repeat inside each part until every part holds one symbol

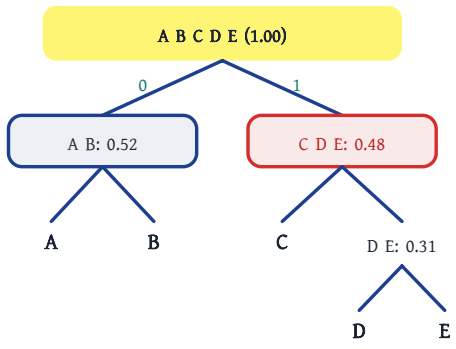
Worked example ✍️: A 0.35, B 0.17, C 0.17, D 0.16, E 0.15

Symbol	A	B	C	D	E
Code	**00**	**01**	**10**	**110**	**111**
l	2	2	2	3	3

Splits: 0.52 | 0.48, then 0.17 | 0.31

$L = 2(0.35 + 0.17 + 0.17) + 3(0.16 + 0.15) =$ **2.31 bits**

$H = 2.233$ bits → $\eta = 2.233/2.31 =$ **96.7%**, $\gamma = 3.3\%$



split as evenly as possible, top-down

Exam trap: Shannon–Fano is **not always optimal**: on this same source Huffman gives 2.30 bits (next sheet).

7 Huffman Coding (bottom-up, optimal)

Definition: D. A. Huffman (MIT, **1952**). Builds the code tree from the leaves up by repeatedly **merging the two least probable** entries. It gives the **minimum average length** of any symbol-by-symbol prefix code.

1. Arrange the probabilities in decreasing order
2. **Merge the two smallest** into one node whose probability is their sum; label the branches 0 and 1
3. Re-sort and repeat until one node of probability 1 remains (M − 1 merges)
4. Read each codeword from the root down to the leaf

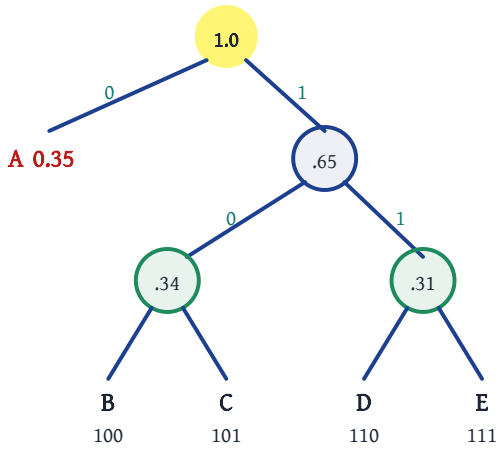
Worked example ✍️: Same source: A 0.35, B 0.17, C 0.17, D 0.16, E 0.15

1. D + E = 0.15 + 0.16 = **0.31**
2. B + C = 0.17 + 0.17 = **0.34**
3. 0.31 + 0.34 = **0.65**
4. 0.65 + A 0.35 = **1.00**

Codes: A = **0**, B = 100, C = 101, D = 110, E = 111

L = 0.35 × 1 + 0.65 × 3 = **2.30 bits**

η = 2.233/2.30 = **97.1%**, γ = 2.9% · Kraft: ½ + 4 × ⅛ = 1 ✓



left branch = 0, right branch = 1

Code for A–E	L (bits)	η	Remark
Fixed length	3	74.4%	$\lceil \log_2 5 \rceil = 3$
Shannon–Fano	2.31	96.7%	top-down split
Huffman	2.30	97.1%	optimal; H = 2.233 is the floor

Memory trick: **Huffman = Hug the two smallest** (bottom-up merge). **Fano = Fold in half** (top-down split).

Exam trap: Ties can give **different codewords** (and code-length variance), but the average length L is always the same. Dyadic source ½, ¼, ⅛, ⅛ → codes 0, 10, 110, 111, L = H = 1.75, η = **100%**.

IT link: Huffman (lossless) is the last stage of *DEFLATE* (ZIP, gzip, PNG = LZ77 + Huffman), of JPEG and of MP3 (their loss comes from earlier stages). Entropy explains why random or already-compressed data (H ≈ 8 bits per byte) cannot be zipped further.

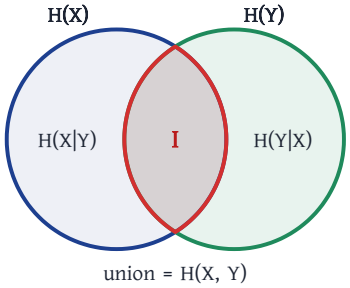
8 Joint & Conditional Entropy, Mutual Information

Joint: $H(X, Y) = H(X) + H(Y|X) = H(Y) + H(X|Y)$

Equivocation $H(X|Y)$: input uncertainty left after seeing the output (lost to noise)

Mutual information (what gets through):
$I(X;Y) = H(X) - H(X|Y)$ $= H(Y) - H(Y|X)$

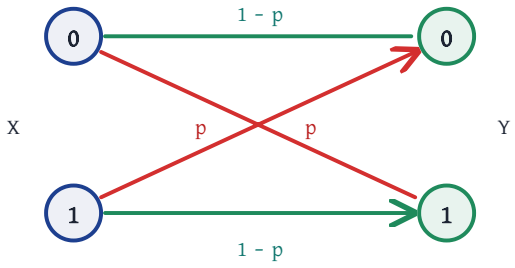
- $I \geq 0$, symmetric · independent X, Y → $I = 0$
- Noiseless channel → $H(X|Y) = 0$, $I = H(X)$



the information diagram

9 Channel Models & Channel Capacity

Definition: A **discrete memoryless channel** (DMC) is described by its transition matrix $P(y_j|x_i)$. Its **capacity** is the largest mutual information over all input distributions: **$C = \max I(X;Y)$** bits per channel use.



binary symmetric channel (BSC), crossover p

BSC capacity (equiprobable input is optimal): **$C = 1 - H(p)$** bits/use

p	0	0.01	0.1	0.2	0.5	1
C	1	0.919	0.531	0.278	0	1

Noiseless, M symbols: $C = \log_2 M$ · erasure channel: $C = 1 - \alpha$

Exam trap: BSC with $p = 1$ is **perfect** ($C = 1$: just invert every bit). The useless one is **$p = 0.5$**.

10 Shannon's Channel Coding Theorem

Shannon's 2nd theorem: If the information rate **$R < C$**, there exists a channel code that makes the **probability of error arbitrarily small**. ($R = C$ is the boundary.) If $R > C$, no code can do so: the error probability stays bounded away from zero.

Memory trick: "**Below C, code and be free; above C, errors you'll see.**" Good codes **exist** (turbo, LDPC, polar come close).

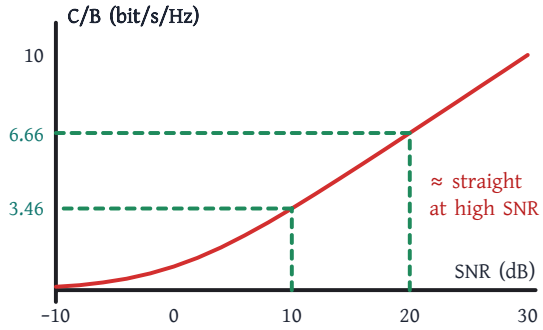
Exam trap: "Error-free" means **arbitrarily low error probability**, not zero errors; the price is **long codes / delay**.

11 Shannon–Hartley Capacity Theorem

Statement: For a band-limited channel of bandwidth B (Hz) with additive white Gaussian noise (**AWGN**) and average signal-to-noise power ratio S/N:

$C = B \log_2(1 + S/N)$ bits/s where $N = N_0 B$ (N_0 = noise power spectral density).

- S/N is a **linear power ratio**, never dB
- C grows **linearly** with B, only **logarithmically** with S/N
- High S/N: +3 dB (×2) adds ≈ 1 bit/s/Hz
- Quick estimate: **$C \approx B \times SNR_{dB}/3$**
- S/N = 0 → C = 0; S/N → ∞ (fixed B) → C → ∞
- An upper limit: real modems stay below it



$C/B = \log_2(1 + S/N)$ against SNR in dB

dB	S/N	$\log_2(1+S/N)$
0	1	1
3	≈ 2	1.58
4.77	3	2
10	10	3.46
11.76	15	**4**
17.99	63	**6**
20	100	6.66
30	**1000**	**9.97**

dB conversions:

Power: $SNR_{dB} = 10 \log_{10}(S/N)$

Voltage: $20 \log_{10}(V_s/V_n)$

Back to linear: $S/N = 10^{SNR_{dB}/10}$

Every +10 dB = ×10 power; +3 dB ≈ ×2

Memory trick: Pick $S/N = 2^k - 1$ (1, 3, 7, 15, 31, 63, 255) and $\log_2(1 + S/N) = k$ exactly: that is how MCQ numbers are built.

Worked examples ✍️:

1. B = 4 kHz, S/N = 15: $C = 4000 \times \log_2 16 = 4000 \times 4 =$ **16 kbit/s**
2. Telephone line B = 3 kHz, SNR = 30 dB → S/N = 1000: $C = 3000 \times \log_2 1001 = 3000 \times 9.97 \approx$ **29.9 kbit/s** (estimate: 3000 × 30/3 = 30 k ✓)
3. B = 1 MHz: 10 dB → 3.46 Mbit/s; 20 dB → 6.66 Mbit/s. B = 10 MHz at 10 dB → **34.6 Mbit/s** (10× B → 10× C)
4. Minimum S/N to carry 20 kbit/s in 5 kHz: $\log_2(1 + S/N) = 4$ → S/N = 15 = **11.76 dB**

Exam trap: Plugging "20 dB" as 20 gives $\log_2 21 = 4.39$ instead of $\log_2 101 = 6.66$. **Convert dB to linear first.**

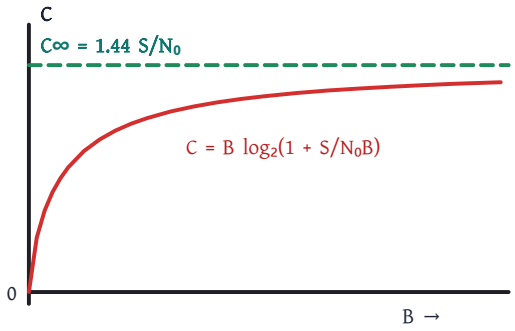
12 Infinite Bandwidth & the Shannon Limit

Fixed S and N_0: wider B lets in more noise ($N = N_0B$). As $B \to \infty$, with $\ln(1 + x) \approx x$:

$C_\infty = (S/N_0) \log_2 e \approx 1.44\ S/N_0$

Capacity **saturates**: infinite B ≠ infinite C.

Put $S = E_b \cdot C$: $E_b/N_0 \geq \ln 2 = 0.693 =$ **−1.59 ≈ −1.6 dB**



capacity flattens out as bandwidth grows

Minimum $E_b/N_0 = (2^{C/B} - 1)/(C/B)$: **0 dB** at 1 bit/s/Hz, 1.76 dB at 2, 5.74 dB at 4.

Exam trap: −1.6 dB limits **E_b/N_0** (energy per bit), not SNR, and needs infinite bandwidth. Below it, reliable communication is impossible at **any** rate.

13 Bandwidth–SNR Trade-off

Same C = 12 kbit/s			
B	S/N	dB	S (× N_0·1 kHz)
3 kHz	15	11.76	45
4 kHz	7	8.45	28
6 kHz	3	4.77	18
12 kHz	1	0	12
∞	→ 0	−∞	8.32 floor

For a fixed C, **bandwidth can be traded for signal power**: more B → less S/N (and less S) needed. Basis of **spread spectrum / CDMA** and wideband FM.

Wide B, low S/N: deep space, CDMA. Narrow B, high S/N: phone modems, high-order QAM.

14 Nyquist Capacity (noiseless channel)

Nyquist bit rate: A noiseless channel of bandwidth B carries at most 2B symbols/s (baud); with M levels: **$C = 2B \log_2 M$** bits/s. Shannon sets the noise ceiling; Nyquist says how many levels reach it.

Worked examples ✍️: ① B = 3 kHz: M = 2 → 6 kbit/s; M = 4 → **12 kbit/s**. ② B = 1 MHz, S/N = 63: Shannon $C = 10^6 \times \log_2 64 =$ **6 Mbit/s**; use 4 Mbit/s for margin → $2 \times 10^6 \times \log_2 M = 4 \times 10^6$ → **M = 4**. ③ 265 kbit/s on a noiseless 20 kHz line: $\log_2 M = 6.625$ → M = 98.7 → use **128** levels.

Memory trick: **Nyquist = No noise, Number of levels** (2B $\log_2$M). **Shannon = S/N** (B $\log_2(1 + S/N)$). Noisy channel: the answer is the **smaller** of the two.

15 Error-Control (Channel) Coding Basics

Idea: The channel coding theorem says reliable transmission needs **controlled redundancy**. An **(n, k) block code** maps k data bits to n-bit codewords: n − k check bits, **code rate r = k/n** (< 1).

	ARQ (Automatic Repeat reQuest)	FEC (Forward Error Correction)
Receiver	detects, asks for retransmission	detects **and corrects** itself
Used in	TCP, Wi-Fi MAC, CRC + retransmit	satellite, deep space, CD/DVD, 4G/5G

Parity check (single-bit)

Even parity: total 1s (data + parity) even. 1011001 has four 1s → even parity bit **0**, odd parity bit **1**. Detects any **odd** number of errors, misses even numbers, **cannot correct** (d_{min} = 2).

Hamming distance

d(a, b) = positions that differ = 1s in a ⊕ b. 10101 ⊕ 11110 = 01011 → **d = 3**. d_{min} = smallest distance between any two codewords.

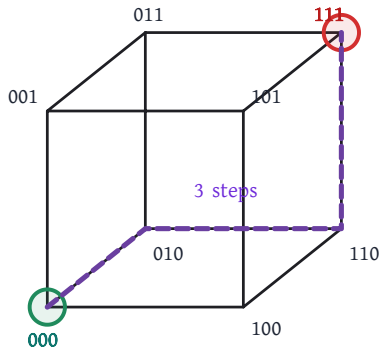
d_{min} rules:

Detect up to **s = d_{min} − 1** errors

Correct up to **t = ⌊(d_{min} − 1)/2⌋** errors

To correct t: $d_{min} \geq 2t + 1$ · detect s: $d_{min} \geq s + 1$

d_{min}	2	3	4	5	7
detect	1	2	3	4	6
correct	0	1	1	2	3



repetition code {000, 111}: d_{min} = 3 corrects 1 error (majority vote)

Hamming code (single-error correcting, d_{min} = 3)

Check bits r: **$2^r \geq m + r + 1$** (m data bits): m = 4 → r = 3, **(7, 4)**; m = 7 (ASCII) → r = 4, (11, 7). Parity bits at positions **1, 2, 4, 8 …**; P1 checks 1, 3, 5, 7 · P2: 2, 3, 6, 7 · P4: 4, 5, 6, 7.

Worked example ✍️: Data 1011 (even parity) → positions 3, 5, 6, 7 = 1, 0, 1, 1. P1 = 1 ⊕ 0 ⊕ 1 = 0 · P2 = 1 ⊕ 1 ⊕ 1 = 1 · P4 = 0 ⊕ 1 ⊕ 1 = 0 → codeword **0110011**.

Received 0110001: checks C4 C2 C1 = 1 1 0 → syndrome **110 = 6** → flip bit 6 → 0110011 ✓

Memory trick: "**Detect d − 1, correct half of that**" (round down).

Exam trap: Parity **cannot correct** even one error; CRC only **detects**.

Night-Before-the-Exam Sheet

Formula sheet:

$I = \log_2(1/p)$ bits · $p = 1 \to I = 0$

$H = \Sigma\ p \log_2(1/p) \cdot 0 \le H \le \log_2 M$

$R = r \cdot H$ bits/s · $H(X^n) = nH$

Kraft $\Sigma\ 2^{-l} \le 1 \cdot H \le L < H + 1$

BSC: $C = 1 - H(p) \cdot p = 0.5 \to C = 0$

$SNR_{dB} = 10 \log_{10}(S/N) \cdot 30$ dB = 1000

Nyquist (noiseless): $C = 2B \log_2 M$

Detect $d_{min} - 1$ · correct $\lfloor (d_{min} - 1)/2 \rfloor$

1 nat = 1.443 bit · 1 Hartley = 3.322 bit

Binary: H(0.5) = 1 bit · H(0.1) = 0.469

$L = \Sigma\ p\ l \cdot \eta = H/L \cdot \gamma = 1 - \eta$

$I(X;Y) = H(X) - H(X|Y) \cdot C = \max I$

$C = B \log_2(1 + S/N)$, S/N linear

$C_\infty = 1.44\ S/N_0 \cdot E_b/N_0 \ge -1.6$ dB

Hamming: $2^r \ge m + r + 1 \cdot$ rate k/n

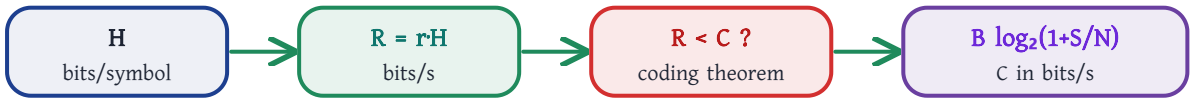
Noisy channel: min(Shannon, Nyquist)

Top 12 one-liners for MCQs

1. Father of information theory: Shannon, 1948
2. Base-2 unit = bit; base e = nat; base 10 = decit
3. p = 1/8 → 3 bits; p = 1 → 0 bits
4. 4 equiprobable symbols → H = 2 bits
5. ½, ¼, ⅛, ⅛ → H = 1.75 bits
6. Huffman (1952): optimal prefix code, merge 2 smallest
7. Shannon–Fano: top-down, not always optimal
8. B = 4 kHz, S/N = 15 → C = 16 kbit/s
9. 30 dB = 1000; 20 dB = 100; 3 dB ≈ 2
10. B → ∞: C → 1.44 S/N_0 (finite)
11. Parity detects odd errors only, corrects none
12. Hamming (7, 4): 3 parity bits, corrects 1

Top traps: dB into the formula without converting · H in bits/symbol vs R in bits/s · baud ≠ bit/s · BSC p = 1 is perfect, p = 0.5 is useless · "error-free" = arbitrarily small error, only for R < C · infinite B ≠ infinite C · Huffman codewords not unique, L is · noisy channel: min(Shannon, Nyquist)

All tricks together: Source coding Shrinks, Channel coding Covers · equal → log, else less · Huffman hugs the two smallest, Fano folds in half · $S/N = 2^k - 1$ gives k · Nyquist = No noise, Number of levels · below C, code and be free · detect d - 1, correct half of that



the four syllabus words in one chain

✓ Practice MCQs (exam pattern, 10 marks)

1. Unit of information for log base e:
(a) bit (b) decit
(c) nat (d) baud

2. Entropy of 8 equiprobable symbols:
(a) 3 bits (b) 8 bits
(c) 1 bit (d) 2.5 bits

3. Information in an event of p = 1/16:
(a) 16 bits (b) 0.0625 bit
(c) 2 bits (d) 4 bits

4. Capacity of a BSC with p = 0.5:
(a) 1 bit/use (b) 0
(c) 0.5 bit/use (d) infinite

5. B = 3 kHz, S/N = 7 (linear). C = ?
(a) 21 kbit/s (b) 24 kbit/s
(c) 9 kbit/s (d) 3 kbit/s

6. 20 dB SNR is a power ratio of:
(a) 20 (b) 200
(c) 10 (d) 100

7. Shannon limit on E_b/N_0:
(a) 0 dB (b) −1.6 dB
(c) 1.6 dB (d) 3 dB

8. d_{min} = 5 corrects up to:
(a) 2 errors (b) 4 errors
(c) 5 errors (d) 1 error

9. Huffman coding gives a ... code:
(a) lossy (b) fixed-length
(c) lossless prefix (d) error-correcting

10. Noiseless 4 kHz, 8 levels: max rate?
(a) 24 kbit/s (b) 12 kbit/s
(c) 32 kbit/s (d) 8 kbit/s

Answer key → 1-c · 2-a · 3-d · 4-b · 5-c · 6-d · 7-b · 8-a · 9-c · 10-a

Written question: A telephone channel has a bandwidth of 3.4 kHz and an SNR of 30 dB. (a) Find its Shannon capacity. (b) What SNR would be needed to carry 40 kbit/s? (c) How many signal levels should a Nyquist-type modem use to stay within capacity? (d) State the spectral efficiency. 8 marks

1. $S/N = 10^{30/10} = 1000$; $C = 3400 \times \log_2 1001 = 3400 \times 9.967 \approx$ **33.9 kbit/s** 3
2. $\log_2(1 + S/N) = 40\,000/3400 = 11.76 \rightarrow S/N = 2^{11.76} - 1 \approx 3479 \rightarrow$ **≈ 35.4 dB** 2
3. $2B \log_2 M \le C \rightarrow \log_2 M \le 33\,900/6800 = 4.98$. M = 32 needs 34.0 kbit/s > C, so use **M = 16** (27.2 kbit/s) 2
4. $C/B \approx 9.97$ bit/s/Hz ($\approx SNR_{dB}/3 = 10$ ✓) 1

Before the exam ✓

- ✓ Can compute I, H and R with units
- ✓ Can build Shannon–Fano and Huffman codes, find L, η, γ
- ✓ Can convert dB ↔ linear and apply Shannon–Hartley
- ✓ Can use d_{min} rules and encode Hamming (7, 4)

Also practise:
① Huffman: 0.4, 0.2, 0.2, 0.1, 0.1
② Derive $C_\infty = 1.44\ S/N_0$
③ BSC capacity, p = 0.1
④ Hamming (7, 4) of 1101